

BAYANKHONGOR, Mongolia

WMO: 442870

Lat: 46.1739N

Lon: 100.7103E

Elev: 1859

StdP: 80.90

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.9	-26.9	-34.5	0.2	-25.7	-32.8	0.2	-24.4	10.2	-18.7	9.1	-14.8	3.3	0	0.702

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	28.2	14.2	26.5	13.5	24.9	12.9	16.6	24.4	15.4	23.0	14.3	21.6	4.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	12.5	19.7	12.5	11.4	18.3	11.4	10.5	17.7	54.0	24.4	50.0	23.1	46.5	21.7	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.5	8.5	-31.2	30.8	2.2	1.3	-32.8	31.8	-34.1	32.5	-35.4	33.2	-37.0	34.1		
(5)			-31.5	17.3	2.2	2.3	-33.1	18.9	-34.4	20.3	-35.6	21.6	-37.2	23.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.4	-17.7	-13.7	-5.7	3.7	9.8	15.8	18.4	16.5	10.2	1.4	-8.3	-15.0		
	DBStd	13.33	4.74	5.36	5.95	4.84	4.50	3.45	3.10	3.00	4.13	4.85	5.55	4.56		
	HDD10.0	3903	858	664	487	193	59	3	0	1	47	268	549	774		
	HDD18.3	6266	1116	897	746	439	266	88	39	72	246	526	799	1032		
	CDD10.0	748	0	0	0	4	53	178	259	203	52	0	0	0		
	CDD18.3	69	0	0	0	0	1	13	40	15	1	0	0	0		
	CDH23.3	717	0	0	0	0	21	168	368	153	7	0	0	0		
	CDH26.7	134	0	0	0	0	2	29	79	24	0	0	0	0		
(14) Wind	WSAvg	3.6	3.2	3.4	3.7	4.1	4.3	3.8	3.4	3.3	3.4	3.2	3.4	3.4		
(15) Precipitation	PrecAvg	209	2	3	5	9	16	37	59	49	17	7	3	2		
	PrecMax	359	9	10	36	37	70	104	152	123	53	53	20	9		
	PrecMin	108	0	0	0	0	0	1	5	2	0	0	0	0		
	PrecStd	74	2	3	7	9	18	28	30	33	12	10	4	3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.1	1.8	11.7	19.5	25.8	29.7	30.9	29.1	24.3	16.4	8.0	-1.6		
		MCWB	-6.3	-3.2	3.2	8.1	11.1	14.3	16.6	14.0	11.4	6.3	1.4	-5.4		
	2%	DB	-6.1	-1.0	8.7	16.9	22.9	27.1	28.8	26.9	21.7	13.9	4.8	-4.2		
		MCWB	-8.7	-5.1	0.8	6.0	10.0	13.5	14.8	13.5	10.1	5.1	-0.7	-7.3		
	5%	DB	-8.6	-3.4	6.0	14.8	20.7	25.3	27.1	25.0	19.7	11.5	2.1	-6.5		
		MCWB	-10.7	-6.7	-0.6	5.0	8.9	12.6	14.1	13.0	9.3	3.7	-2.4	-9.0		
	10%	DB	-10.9	-5.7	3.4	12.6	18.8	23.4	25.4	23.2	17.6	9.2	-0.4	-8.4		
		MCWB	-12.6	-8.3	-2.0	3.9	8.0	11.8	13.5	12.5	8.4	2.4	-4.0	-10.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.1	-3.1	3.6	9.8	13.4	17.2	19.2	17.5	13.4	7.9	1.6	-5.3		
		MCDB	-3.2	1.0	10.6	18.3	23.8	26.1	28.2	24.5	21.6	15.1	7.6	-2.3		
	2%	WB	-8.6	-4.9	1.2	7.0	11.3	15.1	17.1	15.9	11.5	5.7	-0.6	-7.2		
		MCDB	-6.2	-1.3	7.8	15.0	20.7	23.7	25.2	23.0	18.3	12.8	4.4	-4.4		
	5%	WB	-10.7	-6.6	-0.3	5.5	9.9	14.0	15.9	14.7	10.3	4.2	-2.3	-8.9		
		MCDB	-8.7	-3.4	5.6	13.5	18.8	22.2	24.0	21.1	17.7	10.8	1.8	-6.5		
	10%	WB	-12.6	-8.3	-1.8	4.1	8.6	13.0	14.7	13.6	9.1	2.7	-4.0	-10.4		
		MCDB	-11.0	-5.7	3.4	12.0	17.5	20.6	22.4	20.1	16.3	8.6	-0.5	-8.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	12.1	12.5	12.2	12.3	11.8	10.8	11.1	11.8	11.8	10.9	9.8		
		MCDBR	11.8	13.7	14.3	15.0	14.9	14.2	13.3	13.4	13.7	13.9	12.6	11.3		
		MCWBR	9.9	10.7	9.0	8.2	7.4	6.1	4.9	5.3	6.4	8.2	8.7	9.2		
		MCDBR	11.7	13.5	13.7	14.2	13.8	12.3	11.6	10.5	12.3	13.3	12.3	11.2		
	5% WB	MCWBR	9.9	10.6	8.9	8.6	8.0	6.4	5.7	5.7	6.7	8.4	8.8	9.2		
(40) Clear-Sky Solar Irradiance	taub		0.224	0.251	0.293	0.333	0.348	0.359	0.357	0.333	0.301	0.273	0.242	0.225		
	taud		2.402	2.338	2.267	2.231	2.248	2.308	2.361	2.403	2.444	2.448	2.430	2.396		
	Ebn at Noon		912	940	936	922	916	905	902	913	920	902	877	869		
	Edh at Noon		68	90	113	130	133	126	118	108	94	80	66	61		
(44) All-Sky Solar Radiation	RadAvg		1.99	3.08	4.55	5.96	6.83	6.74	6.39	5.80	5.01	3.49	2.20	1.66		
	RadStd		0.16	0.17	0.18	0.26	0.42	0.42	0.33	0.33	0.22	0.16	0.13	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	-2.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	-2.61	N/A	-0.78	-1.03	N/A	N/A	N/A	N/A

Nomenclature: See separate page